

SUMMER INTERNSHIP - 2026

AI Agents for Digital Capability Building in Tata Steel

WEEK-2

AI Use Case Design + Experience Thinking

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Problem Statement

Tata Steel employees across operations, maintenance, logistics, quality, and fabrication roles need continuous skill development to progress in their careers.

Current challenges include:

- * Employees do not know which skills are required for their next role.
- * Training recommendations are generic rather than personalized.
- * Managers lack visibility into team skill gaps.
- * Learning content is scattered across manuals, SOPs, and training documents.
- * Employees spend significant time searching for information.

The objective is to build an AI-powered Learning & Development Assistant that provides personalized learning guidance, role progression support, and instant knowledge access.

Problem / Pain Point	Our Solution Feature
Employees don't know career progression path	AI Career Coach showing current role → target role → readiness score
Employees don't know what skills they lack	Skill Gap Analysis Engine comparing user skills with target role skills
Generic training recommendations	Personalized Course Recommendation Engine based on missing skills
Learning content scattered across manuals and PDFs	AI Knowledge Assistant (RAG) over SOPs, HR policies, safety manuals, NSDC role documents
Employees spend time searching documents	Natural Language Query Interface ("What is chemical spill procedure?")
Managers lack visibility into team learning	Manager Dashboard with team completion, readiness and top skill gaps
Executives lack department-level insights	Executive Dashboard with department analytics and manager comparison
L&D team cannot identify organization-wide skill gaps	L&D Analytics Dashboard showing top skill gaps, course adoption and readiness
Training not aligned with career goals	Role-Based Learning Paths mapped to Tata Steel job roles and NSQF levels
Employees struggle to understand training content	Multilingual Support (If Possible)

Expected Features

Employee Portal

Profile

Career Progression

Skill Gap Analysis

Learning Recommendations

AI Assistant

Learning Progress

L&D Portal

Organization Analytics

Course Effectiveness

Skill Gap Intelligence

Training Planning

AI Assistant

Career Coach

Skill Recommender

Knowledge Assistant

Learning Navigator

This is much stronger than just saying:

AI Coach

AI Trainer

Knowledge Assistant

Executive Portal

Department Analytics

Manager Comparison

Department Readiness

Skill Gap Trends

Manager Portal

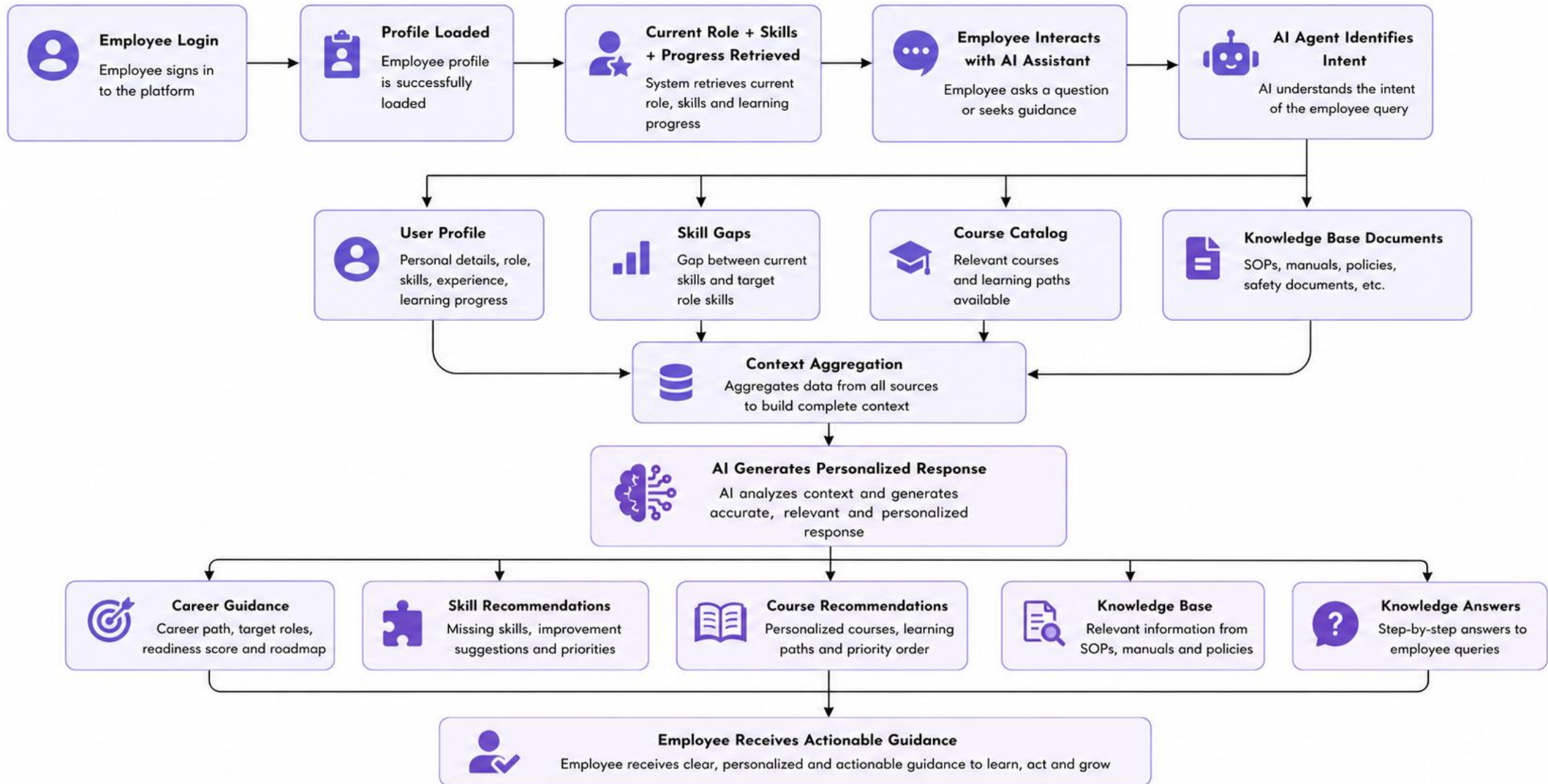
Team Dashboard

Skill Gap Monitoring

Readiness Tracking

Training Progress Monitoring

Complete Design



User Persona

Persona: Shopfloor Employee

Profile:

Works in plant operations or maintenance

Limited technical background

Uses mobile device or kiosk

Needs quick, simple answers

Pain Points:

Doesn't know next career step

Unsure which courses to take

Difficult to search large manuals

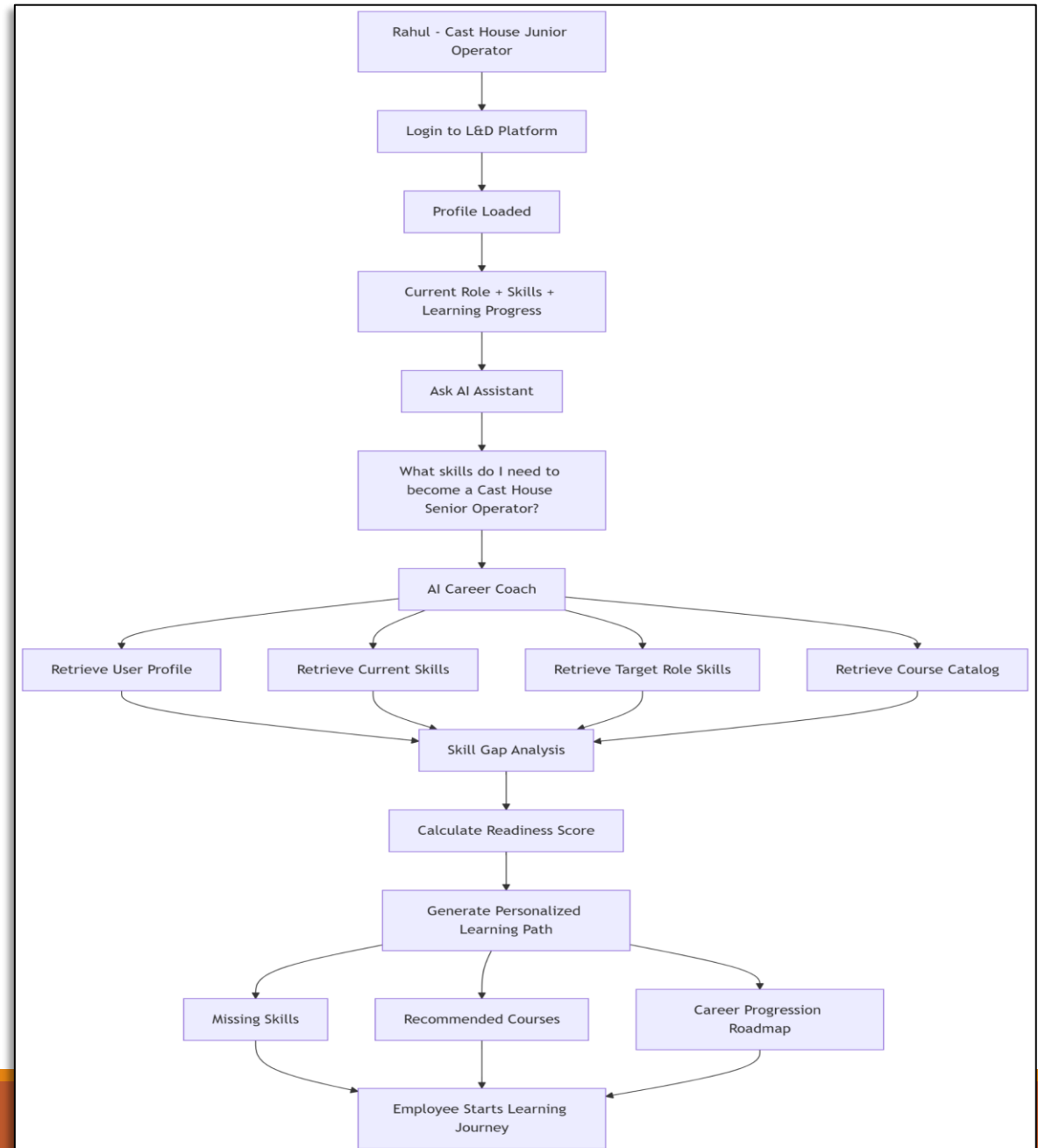
Training information spread across systems

Goals:

Learn required skills

Progress to next role

Find relevant training courses



Sample Prompt and Expected Output for Persona 1

Persona 1: Career Growth Employee

Sample Prompt

Person: "I am currently a Cast House Junior Operator. What do I need to become a Cast House Senior Operator?"

Expected Output

Target Role: Cast House Senior Operator

Current Role: Cast House Junior Operator

Missing Skills:

- Communication Related Skills
- Problem Solving Attitude
- High Concentration Levels
- Operational Practice of Blast Furnace Stoves

Readiness Score: a score between 0-100 (based on experience + current skills)

Recommended Courses: Advanced Cast House Operations, Industrial Communication Skills,
Problem Solving & Root Cause Analysis

Suggested Learning Path:

Complete the recommended courses and gain proficiency in the identified skill gaps to improve readiness for promotion.

Persona 2: Shopfloor Worker Facing Oil Spill

Name: Amit Singh

Role: Fluid Management Operator – Rolling Mills

Experience: 4 Years

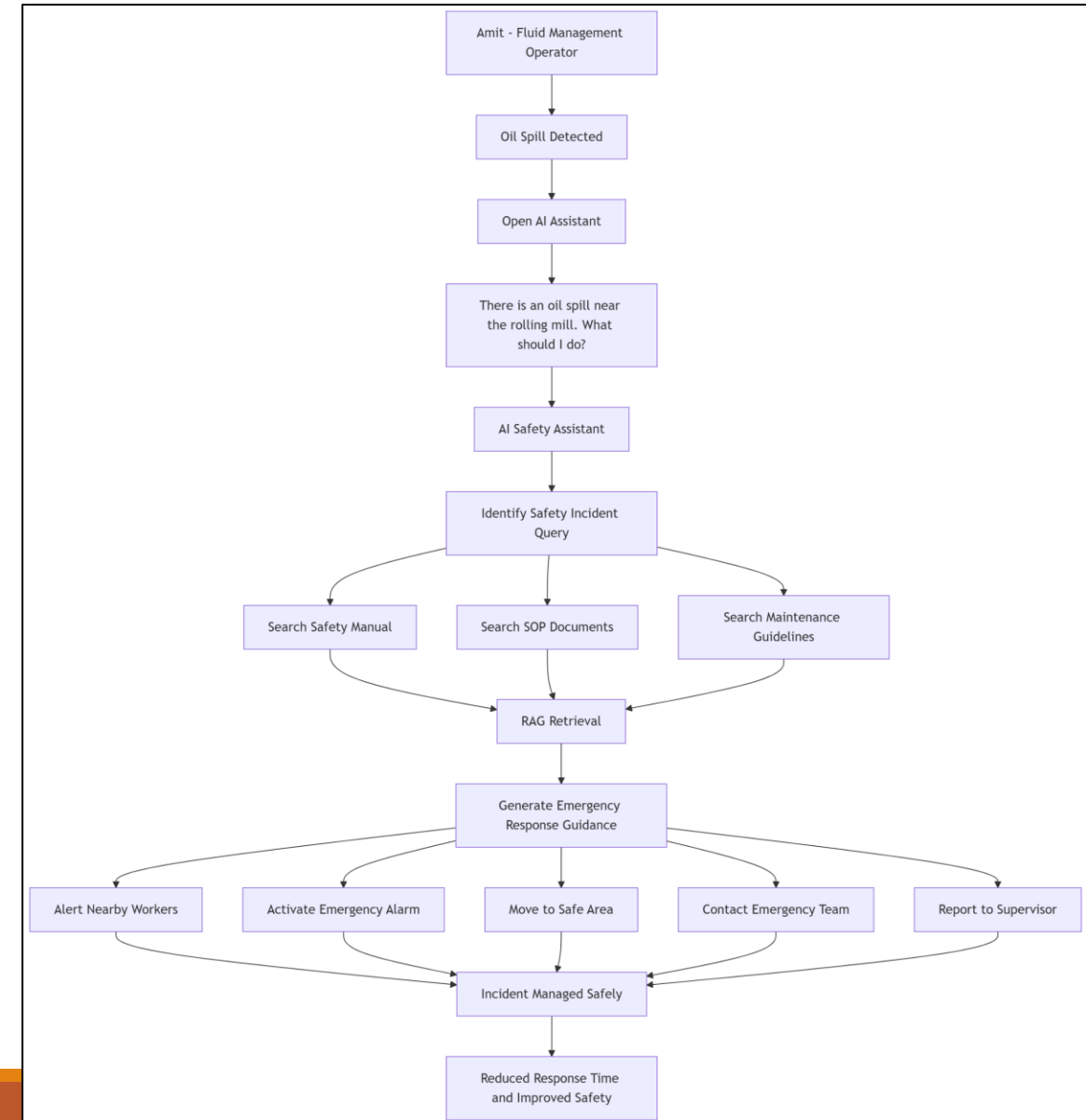
Department: Operations

Goal

Amit wants to respond to the oil spill safely, protect nearby workers, minimize operational risk, and follow the correct procedures.

Pain Points

- Needs immediate guidance during a potentially hazardous situation.
- May not remember the exact emergency response procedure under pressure.
- Cannot spend time searching through lengthy safety manuals.
- Needs accurate, step-by-step instructions to avoid safety violations.
- Requires quick access to emergency contacts and escalation procedures.



Sample Prompt and Expected Output for Persona 2

Shopfloor Worker Facing Oil Spill

Prompt

Amit: "There is a hydraulic oil spill near the rolling mill. What should I do immediately?"

Expected Output

Immediate Safety Actions:

Alert nearby workers immediately.

Activate the emergency alarm.

Isolate the affected area.

Stop nearby equipment if it is safe to do so.

Do not attempt cleanup alone.

Contact the emergency response team.

Follow the approved spill containment procedure.

Report the incident to the supervisor with location and severity details.

Emergency Contact: 1800-TATA-SOS

Source Documents: Safety Manual, Maintenance SOP, Fluid Management Operator Guidelines

Impact

Employee Level:

- * Faster access to knowledge
- * Personalized learning recommendations
- * Clear career progression visibility

Manager Level:

- * Team skill-gap visibility
- * Improved training monitoring
- * Better workforce planning

L&D Level:

- * Department-wide analytics
- * Skill-gap identification
- * Training effectiveness tracking

Expected Outcomes:

- * Reduced time spent searching for information
- * Increased training completion rates
- * Improved workforce readiness
- * Better alignment between training and role requirements

Why This Solution

This solution combines:

- * AI Knowledge Assistant
- * Skill Gap Analysis
- * Personalized Learning Recommendations
- * Career Progression Guidance
- * L&D Analytics

All of the above into a single platform designed specifically for Tata Steel employees.

The system should be usable by shopfloor workers without requiring technical knowledge.

Expected Tech stack and data collection

Datasets Required For AI agent:

Annual Report	- From Tata Portal
Sustainability Report	- From Tata Portal
NSDC PDFs (28 files)	- For Details about Job roles and skillsets required
Safety Manual PDF	-Standard Manuals
HR Policy PDF	- Simulated or From Portal
Leave Policy PDF	- Simulated or From Portal
Maintenance Manual PDF	- Simulated or From Portal
Onboarding PDF	- Simulated or From Portal
Tata Courses CSV	- Made using NSDSC or Pre existing datasets
Equipment Sensor CSV	- Simulated or From Kaggle
Production KPI CSV	- Simulated or From Kaggle
Skills CSV	- Simulated or From Kaggle

Data Layer and Auth

SQLite
PyPDF

JWT Authentication

Frontend

React.js
Vite
Tailwind CSS
Recharts
Axios

AI Layer

Groq (llama-3.1-8b-instant)
LangChain

Backend

FastAPI
Uvicorn

RAG Layer

Chroma DB
Sentence Transformers
(paraphrase-multilingual-MiniLM-L12-v2)
To give user access to Hinglish conversations